Probability

Mind map

Basic terms

- S, events, sample points
- Exclusive, exhaustive: $\sum P = 1$

Classical definition

- $P = \dfrac{n(A)}{n(S)}$
- Treat objects as distinct
- $P(A') = 1 - P(A)$

Bounds

- $0 \le P \le 1$
- $P(A) + P(B) - 1 \le P(AB)$

Odds

- In favour $p : q \Rightarrow \dfrac{p}{p+q}$
- Against $p : q \Rightarrow \dfrac{q}{p+q}$

Dice and coins

- 6^n, 2^n outcomes
- Sum pattern $1, \ldots, 6, \ldots, 1$

Venn diagram

- $P(A \cup B) = P(A) + P(B) - P(AB)$
- Exactly one / two of three
- $S_1 - S_2 + S_3 - \ldots$

Conditional

- $P(A|B) = \dfrac{P(AB)}{P(B)}$
- Given event = new S

Multiplication

- $P(AB) = P(A)P(B|A)$
- ${}^nC_r$ method when order not asked

Independent events

- $P(AB) = P(A)P(B)$
- Three: 4 conditions
- At least one: $1 - \prod q_i$

Repeated trials

- $q^{k-1}p$, $P(X > k) = q^k$
- First player: $\dfrac{p}{1-q^2}$

Total probability

- $\sum P(B_i)P(A|B_i)$
- Tree: add the paths

Bayes' theorem

- Effect given, cause asked
- One path $\div$ all paths
- Extended: next event

Binomial distribution

- ${}^nC_r p^r q^{n-r}$
- Mean np, variance npq
- Least n: $q^n \le 1 - k$

Tree diagram

- Path = product
- Answer = sum of paths

Random variable

- $\sum p_i = 1$
- $\mu = \sum p_i x_i$, $\sigma^2 = \sum p_i x_i^2 - \mu^2$

Geometry and units digit

- Ratio of lengths / areas
- Units digit: last digits only